

Project Plan: SmartHire Recruitment System

Performed by: Lưu Chí Hải, Reviewed by: Nguyễn Gia Quốc Uy, Edited by: Lưu Chí Hải

Planning Baseline: Sprint 2 and Sprint 3 information is based on the team's internal PA planning files and current task assignments. Exact Sprint 1, Sprint 4, and Sprint 5 dates and detailed assignments are team planning baselines created to address the PA2 feedback. They must remain synchronized with the team's Notion task tracker.

1. Introduction

Performed by: Lưu Chí Hải, Reviewed by: Nguyễn Gia Quốc Uy, Edited by: Lưu Chí Hải

SmartHire is an AI-assisted applicant tracking system designed to support recruitment workflows for small and medium-sized enterprises in Vietnam. It centralizes candidate profiles and CVs, job postings, applications, recruitment pipelines, notifications, moderation, and candidate screening in a responsive web application.

The platform serves candidates, company members involved in recruitment, and platform administrators. AI-assisted scoring provides decision-support information and explanations, while authorized recruiters remain responsible for all recruitment decisions.

2. Project Overview

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2.1. Goals

The project aims to:

- provide a centralized recruitment workflow for candidates, company members, and platform administrators;
- reduce fragmented recruitment work performed through disconnected emails and spreadsheets;
- provide candidates with job discovery, application submission, application tracking, and notifications;
- provide authorized company members with job-posting, applicant-review, candidate-scoring, and pipeline-management capabilities;
- provide administrators with employer verification, job moderation, user management, and audit capabilities;
- preserve human control over all candidate-progression and hiring decisions;
- maintain traceability among the Vision Document, functional requirements, use cases, implementation tasks, and tests; and
- deliver the required documents, software builds, reports, and evidence within the project schedule.

2.2. Scope

At a high level, SmartHire covers the following product areas:

1. Authentication, authorization, and company-scoped access control.
2. Account and candidate-profile management.

3. CV upload, parsing, review, and reuse.
4. Job discovery, filtering, saving, sharing, reporting, and application submission.
5. Candidate-side application tracking.
6. Company-scoped job-posting management.
7. Hybrid candidate screening using deterministic matching and AI-assisted semantic analysis.
8. A recruitment pipeline using the canonical application stages defined in the Vision Document.
9. Email and in-app notifications.
10. Employer verification, company-membership management, job moderation, and user management.
11. Backend audit records for security-sensitive and administrative actions.
12. Recruitment analytics and authorized data export as P1 capabilities.

The current project release does not include:

- AI-generated job descriptions;
- AI resume rewriting or resume enhancement;
- fully automated candidate rejection, progression, or hiring;
- semantic AI job recommendations;
- external calendar synchronization;
- payroll;
- employee onboarding; or
- a complete human-resource management suite.

2.3. Delivery Form

SmarterHire is delivered as a responsive web application supporting candidate, recruiter, and administrator workflows.

2.4. Project Constraints

ID	Constraint	Planning Impact
CON-01	The project is completed by a five-member student team under fixed academic deadlines.	Tasks require clear ownership, review responsibility, dates, and acceptance criteria.
CON-02	Project tasks and schedules must be tracked in Notion and remain consistent with this Project Plan.	Assignment or schedule changes must be updated in both locations.
CON-03	Project documents must be written in English and maintained in Markdown and rendered PDF formats.	Documentation tasks include writing, review, conversion, and submission evidence.
CON-04	Source code and project documents must be version-controlled in Git.	Work must preserve commit history and follow the team Git workflow.
CON-05	Required document sections must identify the performer, reviewer, and editor.	A document section is incomplete until its review information is recorded.
CON-06	Recruitment permissions are granted through company-scoped memberships, and company data must remain isolated between tenants.	Backend authorization and testing must verify role, membership, company context, and resource ownership.

ID	Constraint	Planning Impact
CON-07	CVs, contact details, company documents, evaluation notes, and application records are sensitive data.	Secure storage, restricted access, validation, and privacy testing are required.
CON-08	AI scoring is advisory and must not automatically reject, progress, or hire candidates.	Human override and final recruiter control must be preserved.
CON-09	Job matching is rule-based; semantic AI processing is limited to CV-to-job scoring and explanations.	Job-recommendation tasks must not be described as semantic AI recommendations.
CON-10	Supported CV formats are PDF and DOCX, with a maximum file size of 5 MB.	Upload, parsing, scanning, storage, and failure handling must follow these limits.
CON-11	Core operations must continue when the AI service or email service is temporarily unavailable.	Fallback, retry, timeout, logging, and failure handling must be specified and tested.
CON-12	P0 capabilities must be stabilized before P1 analytics and export capabilities.	P1 work may be deferred if P0 acceptance criteria are not satisfied.
CON-13	Spec Kit implementation must preserve its specifications, clarifications, implementation plan, task breakdown, analysis or checklist when used, and source code.	These artifacts are required as implementation evidence.

2.5. Assumptions and Dependencies

The plan assumes that:

- Users have internet access and use a supported modern browser.
- Candidates provide CVs in supported PDF or DOCX formats.
- Recruiters provide accurate job information and valid company-verification documents.
- Administrators review company, membership, and job-posting requests.
- Reviewer assignments follow the Reviewer field recorded for each task in Section 4.
- Notion is updated whenever project scope, ownership, dates, or task status changes.

The project depends on:

- The team Notion task tracker.
- The Git repository.
- The Spec Kit workflow.
- A relational database.
- Secure file storage.
- An external email-delivery service.
- An external AI service.
- A PDF or DOCX file-parsing service.

2.6. Deliverables

The project will produce the following deliverables:

ID	Deliverable	Related Tasks
DEL-01	A responsive SmartHire web application and its source code.	S3-05, S4-01–S4-09, S5-01–S5-03, S5-06, S5-09
DEL-02	A revised Project Plan that addresses PA2 feedback and remains consistent with Notion.	S3-10
DEL-03	A revised Vision Document, with a high-level Product Overview and feature descriptions that include business rationale and primary beneficiaries.	S3-04
DEL-04	One consolidated <code>changes.md</code> file containing separate records for Project Plan and Vision Document revisions.	S3-04, S3-10, S3-12
DEL-05	Spec Kit and specification-driven development artifacts (<code>constitution.md</code> , generated Markdown artifacts, specifications, clarifications, implementation plans, task breakdowns, analysis/checklists, source code).	S2-01, S2-10, S3-05, S3-07
DEL-06	Mermaid use-case models covering the functional requirements in the revised Vision Document.	S3-01, S3-06, S3-08, S3-09
DEL-07	Use-case specifications (IDs, actors, descriptions, preconditions, basic/alternative flows, postconditions).	S3-01, S3-06, S3-08, S3-09
DEL-08	Prototype evidence for the basic and alternative flows of the documented use cases.	S3-01, S3-06, S3-08, S3-09
DEL-09	One full-stack Authentication, Authorization, and Access Control functional group implemented through Spec Kit during Sprint 3.	S3-05, S3-07
DEL-10	A test plan, test cases, test results, and defect records for the implemented system.	S4-10, S5-04, S5-05, S5-07
DEL-11	Weekly Reports, Scrum Meeting records, and Notion task-tracker screenshots.	S1-05, S2-05, S2-15, S3-11, S4-12, S5-11
DEL-12	Individual AI Usage Reports for all team members.	S2-11A–E, S3-03A–E, S4-13A–E, S5-12A–E
DEL-13	User and technical documentation (User Manual, Recruiter User Guide, Administrator Guide, Installation Guide, Deployment Guide, API Documentation, Database Schema Documentation, System Architecture Documentation).	S5-08
DEL-14	Markdown and PDF versions of the required project documents.	S2-16A–C, S2-17, S3-02, S4-11, S5-10

ID	Deliverable	Related Tasks
DEL-15	Git commit-history evidence and the final submission package.	S5-10
DEL-16	Build 1: full-stack Authentication, Authorization, and Access Control increment.	S3-05, S3-07
DEL-17	Build 2: integrated candidate and recruiter recruitment workflows.	S4-01–S4-11
DEL-18	Final Build and demonstration materials.	S5-01–S5-10

Deliverable ownership, reviewers, dates, and acceptance criteria are managed in the sprint task lists rather than assigned globally in this section.

3. Project Organization

Performed by: Lưu Chí Hải, Reviewed by: Nguyễn Gia Quốc Uy, Edited by: Lưu Chí Hải

3.1. Team Structure and Roles

Member	Primary Role	Main Responsibilities
Nguyễn Gia Quốc Uy	Project Manager and Lead Spec Writer	Planning, Vision consolidation, specification consistency, final document integration, and release coordination.
Nguyễn Quốc Thành	AI Integration Lead and Backend Developer	Authentication, AI-assisted scoring, notifications, external-service integration, and backend development.
Lưu Chí Hải	Lead UI/UX Designer and Frontend Developer	Candidate and recruiter interfaces, Project Plan maintenance, Weekly Reports, Notion evidence, and frontend integration.
Nguyễn Minh Khôi	Quality Assurance and Frontend Developer	Test planning, test execution, administrator interfaces, quality review, and defect tracking.
Ngô Quốc Tuấn	Database Architect and Backend Developer	Database schema, migrations, company and job-posting backend functions, data integrity, and backup support.

3.2. Review and Responsibility Rules

- Every task has one accountable owner.
- Reviewers verify the expected output and acceptance criteria before a task is marked **Done**.
- Every project task must be created and tracked in Notion before work begins.
- Source code and document changes must follow the team Git workflow.
- Scope, requirement, schedule, priority, architecture, or technology changes must be recorded in **changes.md**.
- Each member must record AI use on the date it occurs.

3.3. Risk Register

ID	Risk	Probability	Impact	Owner	Mitigation	Contingency
R-01	Cross-tenant data exposure	Medium	Critical	Nguyễn Quốc Thành	Verify authentication, platform role, active company membership, company context, and resource ownership on every protected backend request. Add negative authorization tests.	Block the affected release, disable the vulnerable path, inspect audit records, fix authorization, and repeat tenant-isolation tests.
R-02	CV or personal-data leakage	Medium	Critical	Ngô Quốc Tuấn	Use HTTPS, secure file storage, restricted document access, sanitized file names, malware scanning, and privacy tests.	Revoke exposed access, remove affected test data, rotate credentials when necessary, and conduct a privacy review.
R-03	AI bias or misleading score	Medium	High	Nguyễn Quốc Thành	Keep AI output advisory, display human-readable explanations, prevent automatic rejection, and allow recruiter override.	Disable the unreliable AI result and continue with deterministic matching and human review.
R-04	CV parser failure	Medium	High	Ngô Quốc Tuấn	Preserve the original CV, expose parser status, validate format and size, support retry and timeout handling, and require candidate confirmation of parsed data.	Allow manual profile completion and retry parsing without deleting the original file.
R-05	Email delivery failure	Medium	Medium	Nguyễn Quốc Thành	Separate recruitment transactions from email delivery, record failures, prevent duplicate delivery, and retry	Preserve in-app notifications and expose failed-delivery records for administrator troubleshooting.

ID	Risk	Probability	Impact	Owner	Mitigation	Contingency
					failed delivery at least three times with backoff.	
R-06	Database migration or data loss	Low–Medium	Critical	Ngô Quốc Tuấn	Use ACID transactions, referential integrity, regular backups, documented migrations, a tested rollback procedure, $RTO \leq 60$ minutes, and $RPO \leq 24$ hours.	Stop deployment, roll back the migration, restore the latest verified backup, and investigate before retrying.
R-07	AI-service downtime or rate limits	Medium	High	Nguyễn Quốc Thành	Use asynchronous processing, visible scoring status, provider isolation, and rule-based fallback.	Temporarily disable semantic scoring while unrelated recruitment functions continue.
R-08	Scope creep or unstable P0 capabilities	High	High	Nguyễn Gia Quốc Uy	Prioritize P0 capabilities and defer P1 analytics and export until the core workflow is stable.	Remove P1 work from the active sprint and record the decision in changes.md .
R-09	Notion task-tracker inconsistency	Medium	High	Nguyễn Gia Quốc Uy	Review Notion, the Project Plan, and approved team decisions at the end of each sprint.	Resolve the discrepancy with the team, then update Notion, the Project Plan, and the Weekly Report using the same approved task information.
R-10	Integration or regression failure	Medium	High	Nguyễn Minh Khôi	Require review, automated checks, integration checkpoints, and regression tests.	Revert the breaking change and restore the latest stable build.
R-11	Project Plan or Weekly Report inconsistency	Medium	High	Lưu Chí Hải	Update the Project Plan and Weekly Report whenever approved task information changes	Resolve the discrepancy with the Project Manager and update all affected records consistently.

ID	Risk	Probability	Impact	Owner	Mitigation	Contingency
					and cross-check them against Notion before submission.	

4. Project Plan

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The project follows Scrum and is divided into five sprints. Sprint dates, tasks, owners, reviewers, and statuses must be maintained consistently in this document and in Notion.

4.1. Schedule Summary

Sprint	Date Range	Main Objective	Exit Deliverable
Sprint 1 — PA1	25-05-2026 to 07-06-2026	Establish the team, tools, contract, proposal, and competitor survey.	PA1 project baseline
Sprint 2 — PA2	09-06-2026 to 12-07-2026	Create the Project Plan, Vision Document, Spec Kit initialization, and required reports.	PA2 documentation baseline
Sprint 3 — PA3	17-07-2026 to 26-07-2026	Revise PA2 documents, define use cases and prototypes, and implement Authentication through Spec Kit.	Build 1 and PA3 package
Sprint 4 — PA4	27-07-2026 to 09-08-2026	Implement and integrate the main candidate and recruiter recruitment workflows.	Build 2
Sprint 5 — PA5	10-08-2026 to 23-08-2026	Complete administration, hardening, testing, documentation, final integration, and demonstration.	Final Build

4.2. Sprint 1 Task List

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S1-01	Create the task tracker and GitHub project structure.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	30-05-2026	31-05-2026	The task tracker is available to all members, and the GitHub repository structure is ready for project work.
S1-02	Complete the Existing App Survey for LinkedIn.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	01-06-2026	03-06-2026	The LinkedIn survey documents key features, interface patterns, strengths, weaknesses, and reusable ideas.

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S1-03	Complete the Existing App Survey for TopCV.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	01-06-2026	03-06-2026	The TopCV survey documents key features, interface patterns, strengths, weaknesses, and reusable ideas.
S1-04	Complete the Existing App Survey for VietnamWorks.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	01-06-2026	03-06-2026	The VietnamWorks survey documents key features, interface patterns, strengths, weaknesses, and reusable ideas.
S1-05	Write the Scrum meeting minutes.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	01-06-2026	03-06-2026	The Scrum meeting minutes are completed using the required format and accurately record decisions, progress, and action items.
S1-06	Complete the Team Contract and Project Proposal.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	01-06-2026	06-06-2026	The Team Contract defines the team rules and responsibilities, and the Project Proposal clearly presents the product idea, users, scope, and objectives.

4.3. Sprint 2 Task List

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S2-01	Set up and initialize Spec Kit.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	11-06-2026	17-06-2026	The Spec Kit structure is initialized correctly and is ready for the team to create and manage specifications.
S2-02	Complete the Vision Document: Introduction and Positioning.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	11-06-2026	17-06-2026	The Introduction and Positioning sections clearly describe the product context, problem, target market, and product position.

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S2-03	Complete the Vision Document: Stakeholder and User Description.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	11-06-2026	17-06-2026	The stakeholder and user descriptions identify the relevant groups, their needs, responsibilities, and expected benefits.
S2-04	Complete the Vision Document: Product Overview.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	11-06-2026	17-06-2026	The Product Overview summarizes the product perspective, capabilities, assumptions, dependencies, and operating environment.
S2-05	Prepare the first PA2 meeting minutes and Project Plan.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	11-06-2026	17-06-2026	The first PA2 meeting minutes and the Project Plan are completed using the required structure and contain the agreed scope, schedule, and responsibilities.
S2-06A	Study Spec Kit and maintain a learning log.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	11-06-2026	17-06-2026	The learning log records the Spec Kit topics studied, examples practiced, findings, and remaining questions.
S2-06B	Study Spec Kit and maintain a learning log.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	11-06-2026	17-06-2026	The learning log records the Spec Kit topics studied, examples practiced, findings, and remaining questions.
S2-06C	Study Spec Kit and maintain a learning log.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	11-06-2026	17-06-2026	The learning log records the Spec Kit topics studied, examples practiced, findings, and remaining questions.
S2-06D	Study Spec Kit and maintain a learning log.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	11-06-2026	17-06-2026	The learning log records the Spec Kit topics studied, examples practiced, findings, and remaining questions.

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S2-06E	Study Spec Kit and maintain a learning log.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	11-06-2026	17-06-2026	The learning log records the Spec Kit topics studied, examples practiced, findings, and remaining questions.
S2-07	Refine the detailed briefs for the feature groups.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	18-06-2026	24-06-2026	Each feature group has a clear brief describing its purpose, users, main functions, inputs, outputs, and expected value.
S2-08A	Review, comment on, and provide feedback for the feature groups.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	21-06-2026	23-06-2026	Review comments are documented, conflicts or gaps are identified, and actionable improvements are proposed for each assigned feature group.
S2-08B	Review, comment on, and provide feedback for the feature groups.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	21-06-2026	23-06-2026	Review comments are documented, conflicts or gaps are identified, and actionable improvements are proposed for each assigned feature group.
S2-08C	Review, comment on, and provide feedback for the feature groups.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	21-06-2026	23-06-2026	Review comments are documented, conflicts or gaps are identified, and actionable improvements are proposed for each assigned feature group.
S2-09	Translate the 12 Product Feature groups into English.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	25-06-2026	30-06-2026	All 12 Product Feature group names and descriptions are translated accurately and use consistent English terminology.

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S2-10	Complete constitution.md and move it to the correct Spec Kit location.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	10-07-2026	11-07-2026	The constitution is complete, reflects the team rules, and is stored in the correct Spec Kit directory.
S2-11A	Prepare the AI Usage Report according to the required format.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	25-06-2026	30-06-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S2-11B	Prepare the AI Usage Report according to the required format.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S2-11C	Prepare the AI Usage Report according to the required format.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S2-11D	Prepare the AI Usage Report according to the required format.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S2-11E	Prepare the AI Usage Report according to the required format.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S2-12	Create one Mermaid diagram describing an important workflow.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The Mermaid diagram correctly represents one important end-to-end workflow and is readable

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
						in the Markdown document.
S2-13	Create one Mermaid diagram describing an important workflow.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The Mermaid diagram correctly represents one important end-to-end workflow and is readable in the Markdown document.
S2-14	Complete the Vision Document: Non-Functional Requirements.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The Non-Functional Requirements are measurable and cover the required quality attributes, constraints, and operational expectations.
S2-15	Complete the Weekly Report and meeting minutes according to the required format.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The Weekly Report and meeting minutes contain the required planning, progress, issues, decisions, actions, and review information.
S2-16A	Convert the owner's assigned Spec Kit training or learning document to PDF.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The training document is converted to a readable PDF with correct headings, diagrams, formatting, and no missing content.
S2-16B	Convert the owner's assigned Spec Kit training or learning document to PDF.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The training document is converted to a readable PDF with correct headings, diagrams, formatting, and no missing content.
S2-16C	Convert the owner's assigned Spec Kit training or learning document to PDF.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	25-06-2026	30-06-2026	The training document is converted to a readable PDF with correct headings, diagrams, formatting, and no missing content.

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S2-17	Convert and verify all required PA2 Markdown documents in PDF format.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	30-06-2026	30-06-2026	The Project Plan, Vision Document, Weekly Report, AI Usage Reports, and required Spec Kit documents are available as readable Markdown and rendered PDF files.

4.4. Sprint 3 Task List

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S3-01	Create Diagrams 1 and 2, complete the use-case specifications, and prepare draft prototypes for all use cases assigned to Diagrams 1 and 2.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	20-07-2026	23-07-2026	The diagrams cover all assigned functional requirements; every use case has its ID, actors, preconditions, basic flow, alternative flows, postconditions, and prototype evidence.
S3-02	Finalize the consolidation and standardize the project data.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	24-07-2026	25-07-2026	Project documents and data are consolidated, duplicated or conflicting content is resolved, terminology is consistent, and all required PA3 documents are available in both Markdown and rendered PDF formats.
S3-03A	Prepare the AI Usage Report according to the required format.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	20-07-2026	25-07-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S3-03B	Prepare the AI Usage Report according to the required format.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	20-07-2026	25-07-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S3-03C	Prepare the AI Usage Report according to the required format.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	20-07-2026	25-07-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S3-03D	Prepare the AI Usage Report according to the required format.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	20-07-2026	25-07-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S3-03E	Prepare the AI Usage Report according to the required format.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	20-07-2026	25-07-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S3-04	Revise the Vision Document.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	20-07-2026	25-07-2026	The Vision Document incorporates the PA2 feedback: the Product Overview is rewritten at a high level without implementation-specific details; all 12 feature entries include a business rationale and primary beneficiaries; and the feature table remains consistent with the approved scope and detailed feature descriptions.
S3-05	Implement the Authentication, Authorization, and Access Control functional group using Spec Kit.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	20-07-2026	24-07-2026	Registration, email verification, login, logout and session invalidation, password recovery and reset, JWT storage in HttpOnly, Secure, SameSite cookies, database persistence, UI, API and business logic, and the required Spec Kit artifacts are implemented and traceable to the approved requirements and tasks.
S3-06	Create Diagram 3, complete the use-	Ngô Quốc	Nguyễn Gia Quốc Uy	20-07-	23-07-	The diagram covers all assigned functional

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
	case specifications, and prepare draft prototypes for all use cases assigned to Diagram 3.	Tuấn		2026	2026	requirements; every use case has its ID, actors, preconditions, basic flow, alternative flows, postconditions, and prototype evidence.
S3-07	Provide backup support for the Spec Kit task.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	20-07-2026	24-07-2026	Assigned Spec Kit blockers are supported, resolved, or documented with clear follow-up actions before the due date.
S3-08	Create Diagram 4, complete the use-case specifications, and prepare draft prototypes for all use cases assigned to Diagram 4.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	20-07-2026	23-07-2026	The diagram covers all assigned functional requirements; every use case has its ID, actors, preconditions, basic flow, alternative flows, postconditions, and prototype evidence.
S3-09	Create Diagram 5, complete the use-case specifications, and prepare draft prototypes for all use cases assigned to Diagram 5.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	20-07-2026	23-07-2026	The diagram covers all assigned functional requirements; every use case has its ID, actors, preconditions, basic flow, alternative flows, postconditions, and prototype evidence.
S3-10	Revise the Project Plan.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	20-07-2026	24-07-2026	The Project Plan addresses all PA2 feedback and includes updated constraints, deliverables, risk management, reviewer assignments, specific dates, detailed Sprint 2–5 task lists, Vision alignment, and deliverable-to-task traceability.
S3-11	Write the Weekly Report and capture screenshots of the Notion task tracker.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	20-07-2026	24-07-2026	The Weekly Report records the Scrum Meeting attendance, individual status reports, completed and remaining tasks, issues, decisions, action

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
						items, and meeting summary; the screenshots clearly show the relevant Sprint 3 tasks, owners, dates, and statuses.
S3-12	Update and consolidate changes.md for the Project Plan and Vision Document.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	24-07-2026	25-07-2026	A single changes.md file contains separate Project Plan and Vision Document sections. Each material revision identifies the changed section, revision objective, change summary, reason, responsible owner, reviewer or reviewers, revision date, and revised location.

4.5. Expected Sprint 4 Task List

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S4-01	Finalize the relational database schema and migrations.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	27-07-2026	31-07-2026	Schema and migrations support users, companies, memberships, jobs, applications, stages, files, notifications, scores, and audit records.
S4-02	Implement candidate profile, CV upload, parser status, and parsed-data confirmation backend functions.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	29-07-2026	03-08-2026	Supported files are validated, stored securely, preserved on failure, and available for candidate confirmation.
S4-03	Implement the candidate portal.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	27-07-2026	04-08-2026	Candidates can manage profiles and CVs, discover jobs, apply, and track applications.
S4-04	Implement company-scoped job-posting management backend functions.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	31-07-2026	05-08-2026	Authorized company members can manage only their company's postings.

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S4-05	Implement recruiter job-posting and applicant-review interfaces.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	01-08-2026	07-08-2026	Recruiters can manage postings and review permitted applicant data.
S4-06	Implement backend pipeline transitions and stage history using the nine canonical stages.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	02-08-2026	07-08-2026	Transitions are transactional, audited, and restricted to the active company context.
S4-07	Implement the recruitment pipeline Kanban interface.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	03-08-2026	08-08-2026	Authorized users can view and update stages with clear success and failure states.
S4-08	Implement hybrid candidate scoring and rule-based fallback.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	28-07-2026	07-08-2026	Scoring status is separate from pipeline status, explanations are displayed, and AI never changes recruitment decisions automatically.
S4-09	Implement email and in-app notifications.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	01-08-2026	08-08-2026	Recruitment actions complete independently of email delivery; failures are logged and retried.
S4-10	Execute Build 2 functional, integration, authorization, and regression tests.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	03-08-2026	09-08-2026	The test plan and test cases are documented before execution; test results and identified defects are recorded; and the core candidate and recruiter workflows pass with no unresolved critical security or data-integrity defect.
S4-11	Consolidate Build 2 documentation and conduct the Sprint Review.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	08-08-2026	09-08-2026	Scope, evidence, known defects, deferred work, and Sprint Review decisions are recorded consistently, and all required Sprint 4 documents are available in both Markdown and rendered PDF formats.

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S4-12	Write the Weekly Report and capture screenshots of the Notion task tracker.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	27-07-2026	09-08-2026	The Weekly Report includes Sprint Planning, Scrum Meeting evidence, Sprint Review evidence, progress, issues, decisions, and action items; the screenshots clearly show the relevant Sprint 4 tasks, owners, dates, and statuses.
S4-13A	Prepare the AI Usage Report according to the required format.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	27-07-2026	09-08-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S4-13B	Prepare the AI Usage Report according to the required format.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	27-07-2026	09-08-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S4-13C	Prepare the AI Usage Report according to the required format.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	27-07-2026	09-08-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S4-13D	Prepare the AI Usage Report according to the required format.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	27-07-2026	09-08-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S4-13E	Prepare the AI Usage Report according to the required format.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	27-07-2026	09-08-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.

4.6. Expected Sprint 5 Task List

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
S5-01	Implement company verification, membership approval, role management, and membership lifecycle backend functions.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	10-08-2026	15-08-2026	Company access requires the approved membership workflow and all changes are audited.
S5-02	Implement administrator verification, moderation, user-management, and enforcement interfaces.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	10-08-2026	16-08-2026	Administrators can perform permitted actions with recorded decisions and reasons.
S5-03	Harden authentication sessions, secure-cookie handling, and company-scoped authorization.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	10-08-2026	16-08-2026	Protected requests verify authentication, platform role, membership, company scope, and ownership.
S5-04	Execute cross-tenant, CV-access, privacy, authentication, and authorization tests.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	14-08-2026	19-08-2026	Negative tests confirm that unauthorized requests do not expose protected data.
S5-05	Execute AI, parser, email, database, retry, timeout, and recovery tests.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	15-08-2026	20-08-2026	Core operations remain usable and stored data remains consistent during failures.
S5-06	Implement P1 recruitment analytics and authorized export after P0 readiness approval.	Ngô Quốc Tuấn	Nguyễn Gia Quốc Uy	16-08-2026	20-08-2026	The feature is delivered only if P0 is stable and exported data respects authorization.
S5-07	Execute performance, compatibility, accessibility, usability, regression, and end-to-end tests.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	17-08-2026	22-08-2026	Test evidence covers measurable Vision targets and all P0 workflows.
S5-08	Prepare user and technical	Nguyễn Gia	Nguyễn Quốc Thành,	12-08-	22-08-	Manuals, guides, API, database, deployment, and

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
	documentation.	Quốc Uy	Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	2026	2026	architecture documents match the released system.
S5-09	Integrate final frontend and backend changes and resolve release-blocking defects.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	18-08-2026	22-08-2026	The final build is reproducible and contains no unresolved critical defect.
S5-10	Consolidate the final submission and demonstration package.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	21-08-2026	23-08-2026	Source code, documents, PDFs, tests, reports, Git evidence, changes.md , and demo materials are complete.
S5-11	Write the Weekly Report and capture screenshots of the Notion task tracker.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	10-08-2026	23-08-2026	The Weekly Report includes Sprint Planning, Scrum Meeting evidence, Sprint Review and retrospective evidence, progress, issues, decisions, and action items; the screenshots clearly show the relevant Sprint 5 tasks, owners, dates, and statuses.
S5-12A	Prepare the AI Usage Report according to the required format.	Nguyễn Gia Quốc Uy	Nguyễn Quốc Thành, Ngô Quốc Tuấn, Lưu Chí Hải, Nguyễn Minh Khôi	10-08-2026	23-08-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S5-12B	Prepare the AI Usage Report according to the required format.	Nguyễn Quốc Thành	Nguyễn Gia Quốc Uy	10-08-2026	23-08-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S5-12C	Prepare the AI Usage Report according to	Ngô Quốc	Nguyễn Gia Quốc Uy	10-08-	23-08-	The report records the AI tool, version, usage date,

ID	Task and Expected Output	Owner	Reviewer	Start	Due	Acceptance Criteria
	the required format.	Tuấn		2026	2026	prompts, purpose, generated content, verification, and individual contribution.
S5-12D	Prepare the AI Usage Report according to the required format.	Lưu Chí Hải	Nguyễn Gia Quốc Uy	10-08-2026	23-08-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.
S5-12E	Prepare the AI Usage Report according to the required format.	Nguyễn Minh Khôi	Nguyễn Gia Quốc Uy	10-08-2026	23-08-2026	The report records the AI tool, version, usage date, prompts, purpose, generated content, verification, and individual contribution.

4.7. Build Plan

Build	Target Date	Included Scope	Exit Criteria
PA2 Documentation Baseline	12-07-2026	Project Plan, Vision Document, Spec Kit initialization, workflows, NFRs, Weekly Report, and AI Usage Reports.	Required documents and evidence are complete and consistent with Notion.
Build 1	26-07-2026	Revised Vision and Project Plan, use-case models, specifications, prototypes, and full-stack Authentication implementation.	PA3 deliverables are reviewed and Authentication is demonstrable.
Build 2	09-08-2026	Candidate portal, recruiter job management, CV handling, application tracking, nine-stage pipeline, hybrid scoring, and notifications.	Main candidate and recruiter workflows are integrated with no critical access-control or data-integrity defect.
Final Build	23-08-2026	Administration, company membership, moderation, security hardening, failure handling, approved P1 scope, testing, documentation, and demo package.	All P0 acceptance tests pass, release-blocking defects are resolved, and submission materials are complete.

4.8. Definition of Done

A task is considered complete only when:

- 1. It is recorded and updated in Notion.

2. Its expected output is committed to the agreed Git repository structure.
3. Its acceptance criteria are satisfied.
4. The assigned reviewer or reviewers approve it.
5. Required test, prototype, meeting, or report evidence is recorded.
6. Related documents and traceability references are updated.
7. Required Markdown and PDF outputs render correctly.
8. Material scope, schedule, requirement, priority, or technology changes are recorded in `changes.md`.

5. Alignment with the Revised Vision

Performed by: Lưu Chí Hải, Reviewed by: Nguyễn Gia Quốc Uy, Edited by: Lưu Chí Hải

Revised Vision Decision	Project Plan Response
SmartHire targets Vietnamese SMEs.	The Introduction and project goals use the same target-product context.
Product Overview must remain high-level.	S3-04 revises the Vision Document Product Overview to remove implementation-specific details and retain only the product purpose, target users, high-level capabilities, product boundaries, assumptions, and dependencies.
Every feature must include business rationale and beneficiaries.	S3-04 requires all 12 feature entries to include a business rationale and primary beneficiaries and to remain consistent with the detailed feature descriptions and approved scope.
AI-generated job descriptions are removed.	No sprint task or build includes AI job-description generation.
AI resume rewriting is removed.	CV work is limited to upload, parsing, confirmation, reuse, and scoring input.
The recruitment pipeline uses nine canonical stages.	Pipeline backend, Kanban, application tracking, specifications, and tests use the same stages.
A base user retains candidate identity and receives company-scoped permissions through memberships.	Authentication and authorization tasks use the membership model rather than a separate recruiter account.
Authentication tokens use HttpOnly, Secure, SameSite cookies.	Sprint 3 Authentication and Sprint 5 hardening tasks require secure-cookie handling.
Job matching is rule-based.	Semantic AI work is limited to CV scoring and explanations.
AI scoring is advisory.	Constraints, risks, scoring tasks, and tests preserve recruiter override and prevent automated decisions.
Analytics and export are P1.	Sprint 5 schedules them only after P0 readiness approval and permits deferral.
Personal data and CVs require protection.	Constraints, risk mitigations, file-handling tasks, and security tests address protection.

Revised Vision Decision	Project Plan Response
AI, email, parser, and database failures must be handled.	Sprint 4 and Sprint 5 include fallback, retry, timeout, logging, and recovery work.

6. PA2 Feedback Traceability

Performed by: Lưu Chí Hải, Reviewed by: Nguyễn Gia Quốc Uy, Edited by: Lưu Chí Hải

PA2 Feedback	Location Addressed
Sprint 2 lacked reviewers.	Section 4.3 records an explicit reviewer for every Sprint 2 task.
Sprint 2 task assignments were not detailed enough.	Section 4.3 includes expected outputs, owners, reviewers, dates, and acceptance criteria.
Sprints 3–5 lacked task lists.	Sections 4.4, 4.5, and 4.6 provide complete task lists.
The schedule lacked specific dates.	Section 4.1 provides sprint ranges and the task tables provide start and due dates.
Project Overview lacked constraints.	Section 2.4 defines project, review, authorization, data, AI, file, service, priority, and Spec Kit constraints.
Deliverables were incomplete.	Section 2.6 includes software, documents, specifications, prototypes, tests, reports, guides, evidence, builds, and demo materials.
The risk register needed six additional risks.	Section 3.3 includes all six requested risks with mitigations and contingencies.
The Project Plan did not reflect important Vision changes.	Section 5 maps Vision decisions to project scope, constraints, tasks, risks, and tests.
Feature descriptions needed business rationale and beneficiaries.	S3-04 requires all 12 feature entries to include a business rationale and primary beneficiaries, with consistency checked against the detailed feature descriptions and approved scope.
Product Overview was too detailed.	S3-04 requires the Vision Document Product Overview to be rewritten at a high level and removes implementation-specific details.

7. Revision History

Date	Section	Change Introduced	Reason
25-07-2026	Project Overview	Added complete constraints, scope boundaries, assumptions, dependencies, and deliverables.	Address PA2 feedback and align with the revised Vision.
25-07-2026	Reviewer Assignments	Added explicit reviewer assignments to Sprint 2 and Sprint 3 tasks.	Address the PA2 feedback that Sprint 2 tasks lacked reviewers.

Date	Section	Change Introduced	Reason
25-07-2026	Schedule	Added planning date ranges for all five sprints and task-level dates.	Address the missing-date feedback.
25-07-2026	Sprint Task Lists	Added detailed Sprint 2–5 tasks with outputs and acceptance criteria.	Address incomplete Sprint 2 tasks and missing Sprint 3–5 lists.
25-07-2026	Deliverables	Expanded software, documentation, testing, evidence, build, and guide deliverables.	Address incomplete deliverables.
25-07-2026	Risk Register	Added the requested risks with mitigations and contingencies.	Address the PA2 risk feedback.
25-07-2026	Vision Alignment	Added scope, authorization, AI, pipeline, security, and priority mappings.	Reflect confirmed changes in the revised Vision Document.
25-07-2026	Deliverable Traceability	Corrected task references for <code>changes.md</code> , AI Usage Reports, Markdown/PDF outputs, and software builds.	Keep deliverable mappings consistent with the revised task IDs.
25-07-2026	Acceptance Criteria	Corrected the criteria for the Vision revision, Spec Kit implementation, document consolidation, and Sprint 3–5 reporting evidence.	Remove inconsistencies introduced by recent task updates.
25-07-2026	Vision Feedback Traceability	Corrected the mapping of the Vision Product Overview and feature-description feedback to S3-04.	Ensure each PA2 Vision comment points to the actual Vision-revision task.